

01MAT102052004 B

A

Pages: 3

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Second Semester B.Tech Degree Examination July 2021

Course Code: MAT102

Course Name: VECTOR CALCULUS, DIFFERENTIAL EQUATIONS AND TRANSFORMS

(2019 Scheme)

AN Session

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

- 1 Find the velocity and acceleration of a particle whose position vector is given by
$$\vec{r}(t) = e^t \vec{i} + e^{-t} \vec{j} \quad \text{at } t = 0. \quad (3)$$
- 2 If C is the unit circle $x^2 + y^2 = 1$ oriented counter clockwise, find $\int_C x dx + y dy$. (3)
- 3 Determine the sources and sinks of the vector field
$$\vec{f}(x, y, z) = 2(x^3 - 2x) \vec{i} + 2(y^3 - 2y) \vec{j} + 2(z^3 - 2z) \vec{k} . \quad (3)$$
- 4 Evaluate $\iint_{\sigma} xz dS$ where σ is the part of the plane $x + y + z = 1$ that lies in the first octant. (3)
- 5 Solve the initial value problem $y'' + y' - 2y = 0$, $y(0) = 4$, $y'(0) = -5$. (3)
- 6 Solve $y'''' + 10y'' + 9y = 0$. (3)
- 7 Find the Laplace transform of $e^{-3t} \cos 2t$. (3)
- 8 Find the inverse Laplace Transform of $\frac{1}{(s+a)(s+b)}$ (3)
- 9 Find the Fourier cosine transform of the function $f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$ (3)
- 10 Using Fourier sine integral, show that
$$\int_0^{\infty} \frac{1 - \cos \pi w}{w} \sin x w dw = \begin{cases} \frac{\pi}{2} & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases} \quad (3)$$

01MAT102052004 B

PART B

Answer one full question from each module, each question carries 14 marks

Module-I

- 11 a) Find the directional derivative of $\phi(x, y, z) = x^3z - yx^2 + z^2$ at $(1, 1, 1)$ in the direction of $\vec{a} = 2\vec{i} - \vec{j} + 2\vec{k}$. Also find the maximum directional derivative. (7)
- b) Show that the vector field $\vec{f}(x, y) = xy^2\vec{i} + x^2y\vec{j}$ is conservative and find ϕ such that $\vec{f} = \nabla\phi$. Hence evaluate $\int_{(1,2)}^{(2,4)} xy^2 dx + x^2y dy$. (7)
- 12 a) Find the parametric equation of the tangent line to the graph $\vec{r}(t) = t^2\vec{i} - \frac{1}{t+3}\vec{j} + (4-t^2)\vec{k}$ at $(4, -1, 0)$. (7)
- b) Using line integral evaluate $\int_C x^2y dx + x dy$ where C is the triangular path connecting $(0, 0)$, $(1, 0)$ and $(1, 2)$ in the positive direction. (7)

Module-II

- 13 a) Use Green's theorem to evaluate $\int_C \log(1+y) dx - \frac{xy}{1+y} dy$ where C is the triangle with vertices $(0, 0)$, $(2, 0)$ and $(0, 4)$. (7)
- b) Use Divergence theorem to find the outward flux of the vector field $\vec{F} = (2x + y^2)\vec{i} + xy\vec{j} + (xy - 2z)\vec{k}$ across the surface of the tetrahedron bounded by $x + y + z = 2$ and the coordinate planes. (7)
- 14 a) Find the flux of the vector field $\vec{F}(x, y, z) = x\vec{k}$ across the surface, the portion of the paraboloid $z = x^2 + y^2$ below the plane $z = 2y$ oriented by downward unit normal. (7)
- b) Use Stokes theorem evaluate $\oint_C \vec{f} \cdot d\vec{r}$ where $\vec{f}(x, y, z) = (z - y)\vec{i} + (z + x)\vec{j} - (x + y)\vec{k}$ and C is the boundary of the paraboloid $z = 9 - x^2 - y^2$ above the XY-plane with upward orientation. (7)

Module-III

- 15 a) Solve using the method of undetermined coefficients $y'' - 4y' - 5y = 4\cos 2x$. (7)
- b) Solve using the Method of variation of parameters $y'' + y = \operatorname{cosec} x$ (7)

01MAT102052004 B

- 16 a) Solve using the method of undetermined coefficients (7)
 $y'' - 7y' + 12y = e^{2x}$, $y(0) = 1$, $y'(0) = 2$.
- b) Solve the boundary value problem $x^2 y'' - 3xy' + 4y = 0$, $y(1) = 0$, $y'(1) = 1$ (7)

Module-IV

- 17 a) Using convolution theorem find the Laplace inverse of $\frac{2}{(s^2 + 1)(s^2 + 25)}$ (7)
- b) Using Laplace Transform solve $y'' + 2y' + y = e^{-t}$ $y(0) = 0$, $y'(0) = 1$ (7)
- 18 a) Express in terms of unit step function and hence find the Laplace Transform of (7)
$$f(t) = \begin{cases} t^2, & \text{if } 0 < t < 2 \\ t-1 & \text{if } 2 < t < 3 \\ 7 & \text{if } t > 3 \end{cases}$$
- b) Find the inverse Laplace Transform of $\frac{s^2 + 2}{s(s^2 + 9)}$ (7)

Module-V

- 19 a) Find the Fourier Transform of $f(x) = \begin{cases} |x| & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$ (7)
- b) Find the Fourier integral of $f(x) = \begin{cases} \pi - x & \text{if } 0 < x < \pi \\ 0 & \text{otherwise} \end{cases}$ (7)
- 20 a) Find the Fourier Sine Transform of $f(x) = \begin{cases} x & \text{if } 0 < x < 1 \\ 3-x & \text{if } 1 < x < 3 \\ 0 & \text{if } x > 3 \end{cases}$ (7)
- b) Represent $f(x) = e^{-kx}$ $x > 0$, $k > 0$ as a Fourier Cosine integral (7)
